higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

T940**(E)**(A9)T

NATIONAL CERTIFICATE

LOGIC SYSTEMS N4

(8080284)

9 April 2019 (X-Paper)
09:00–12:00

This question paper consists of 5 pages.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
LOGIC SYSTEMS N4
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. ALL the sketches and diagrams must be large, clear and neat.
5. Start each question on a NEW page.
6. Keep questions and subsections of questions together.
7. Questions must be answered in BLUE or BLACK ink and all drawings should be done in PENCIL.
8. ALL calculations must be shown.
9. Calculations and calculated answers must be given in THREE fractional radix spaces, for example $10{,}101_2$.
10. Write neatly and legibly.

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QUESTION 1

1.1 Give the symbols of the octal number system. (1)

1.2 What is the decimal value of a binary 1 which is in the third position to the right of the radix comma? (1)

1.3 Determine the two's complement of 100101_2. (2)

1.4 Give the Boolean expressions for De Morgan's TWO theorems. (2)

1.5 Simplify the following Boolean expressions:

1.5.1 $A\bar{A}+A=$

1.5.2 $A+ABCD+AC$

1.5.3 $A\bar{B}+A$

1.5.4 $A+B+\bar{A}$

(4 × 1) (4)

1.6 Determine the simplest Boolean expression for the following Karnaugh maps. Redraw each Karnaugh map and show the groupings.

1.6.1

CD \ AB	00	01	11	10
00			1	1
01			1	1
11				1
10		1	1	1

1.6.2

CD \ AB	00	01	11	10
00		1	1	
01		1	1	
11	1		1	1
10	1			1

1.6.3

CD \ AB	00	01	11	10
00	1	1	1	1
01	1	1	1	1
11	1	1	1	1
10	1	1	1	1

(3 × 2) (6)

1.7 Draw the truth table for a JK master/slave flip-flop. (4)

[20]

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QUESTION 2

2.1 Convert each of the following numbers to their BINARY equivalents and then complete the calculations in the binary number system. Show all conversions.

2.1.1 $122,3_8$ -$132,C_{16}$
Use two's complement subtraction. Convert the final answer to the decimal numbering system.

2.1.2 $36,4_8 \times 6,25_{10}$
Convert the final answer to the hexadecimal system.

2.1.3 $54,6_8 \div 6,8_{16}$
Convert the final answer to the octal system.

(3 × 6) (18)

2.2 Convert $18,65_{10}$ to the 8421 BCD code. (4)

2.3 Convert $0001011100011,0110_2$ to the Excess 3 code. (3)
[25]

QUESTION 3

3.1 Draw the circuit diagram of each of the following Boolean expressions. Do NOT change or simplify the expressions:

3.1.1 $F=(A+B)\,\overline{(A+B+\bar{C})(\bar{B}+C)}$

3.1.2 $F=\overline{(\overline{AB}+\bar{A}\bar{B}\bar{C})}+B\bar{C}$

3.1.3 $F=\overline{(A\oplus\overline{B\oplus C})\,(A+\bar{B}+C)}$

(3 × 5) (15)

3.2 Simplify the following Boolean expressions by using Boolean laws. Draw the simplified circuit diagrams by using NAND gates only.

3.2.1 $F=(A+B)\,[\bar{A}(\bar{B}+\bar{C})]+\bar{A}C+A\bar{B}$

3.2.2 $F=\overline{(AB+\bar{A}BC)(A\bar{B}C+\bar{A}B\bar{C})+(AB+A\bar{B}C)}$

(2 × 5) (10)
[25]

QUESTION 4

With the aid of Karnaugh maps, design a circuit that will convert the Gray code with four variables (A, B, C, D) to binary. Simplify with the aid of Boolean algebra so that the circuit consists of the minimum number of gates. **[20]**

QUESTION 5

Make a neat, labelled sketch of a D-type master/slave flip-flop which consists of NAND gates only. **[10]**

TOTAL: 100

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